

Firmware — sentinel node

Owner: *TBD*

Depends on: `docs/PROTOCOL.md`, `hardware/` AFE characterisation

Deliverable for M3: three dev boards gossiping over real LoRa, quorum firing correctly.

Do not wait for the PCB. Start on dev boards today: an STM32L4 Nucleo or ESP32-S3 devkit plus an SX1262 breakout gets you 80% of the firmware done before the board exists.

1. Target

	Choice	Notes
MCU	STM32L432KC, or ESP32-S3	STM32 for the power budget; ESP32 if the team already knows it
Radio	SX1262, 865–867 MHz	India ISM band. Not 915 MHz — that is

	Choice	Notes
		unlicensed elsewhere, not here.
Toolchain	PlatformIO	one command builds for both targets
RTOS	none, or FreeRTOS if the mesh needs it	a superloop with a timer ISR is enough and easier to reason about

2. Structure

```

firmware/
├─ platformio.ini
├─ src/
│   ├─ main.c
│   ├─ adc.c           DMA sampling, 10-
cycle RMS envelope
│   ├─ baseline.c      EWMA,  $\tau \approx 10$  min
│   └─ detector.c      SUSPECT / RECOVERED
state machine
│   └─ arbiter.c       C port of backend
decide() – must match vectors
│   └─ mesh.c          LoRa gossip,
slotting, quorum collection

```

```

|   |— radio.c      SX1262 driver
wrapper
|   |— power.c      sleep scheduling,
battery ADC
|   |— proto.c      message pack/unpack
per PROTOCOL.md
|— test/            runs arbiter.c
against docs/vectors/ on host

```

3. Signal chain

```

probe → AFE → ADC (DMA, 1 kHz) → 10-cycle
window (200 ms)
      → RMS → EWMA baseline →
deviation_pct → detector state machine

```

- Sample at 1 kHz — 20 samples per 50 Hz cycle, comfortably above Nyquist with margin for harmonics.
- RMS over exactly 10 cycles (200 ms) so a whole number of cycles is always integrated; partial cycles produce a beat artefact that looks like a fault.
- **Baseline must not adapt fast enough to follow a break.** $\tau \approx 10$ min tracks temperature and load drift while a genuine collapse stays far below baseline indefinitely. If τ is too short, the node "forgets" the fault and silently returns to NORMAL

with a broken wire on the ground. This is the single most dangerous parameter in the firmware.

- Freeze baseline adaptation entirely while in SUSPECT or CONFIRMED.

4. Detector state machine

```
NORMAL —deviation < -60% for 5
consecutive windows—► SUSPECT
SUSPECT —deviation > -20%—► RECOVERED
—2 s—► NORMAL
SUSPECT —quorum met—► CONFIRMED
CONFIRMED —manual reset or RESTORE
command—► NORMAL
```

Thresholds live in `ArbiterConfig` (PROTOCOL.md §5) and must be runtime-configurable, not compile-time constants — the utility will want to tune per feeder.

5. LoRa mesh

- **Slotted TDMA.** Each node gets a transmit slot derived from `node_id % N`. Pure ALOHA collides badly the moment several nodes go SUSPECT at once — which is exactly when you need the messages through.
- **Time sync** from the gateway beacon, once per minute. Cheap GPS PPS is an option if drift proves

problematic.

- **Priority path:** a SUSPECT assertion pre-empts routine telemetry and transmits immediately, out of slot. Latency budget for the whole gossip round is ~800 ms.
- **Packet security:** 4-byte node ID, rolling `seq`, CRC, and an AES-128-CCM MAC with a per-feeder key. An unauthenticated radio that can trigger a feeder isolation is a vulnerability a judge will find.
- **Range:** SF9, BW 125 kHz, CR 4/5 as the starting point. Measure actual span range in the field and adjust; higher SF buys range but costs airtime, and airtime is your latency budget.

6. Power

Target average current < 5 mA.

- Sleep between sampling windows. Sampling duty cycle around 20%.
- Radio RX windows scheduled, not continuous listen.
- Wake on the priority path immediately when local detector asserts.
- Budget: 6 V 1 W panel + 1 × 18650 LiFePO4, sized for 5 days of monsoon overcast. Show this calculation in the PPT.

7. Task list

Phase 1 — dev board (start now)

- ☐ PlatformIO project, blink, UART logging
- ☐ SX1262 driver, two boards exchanging a packet
- ☐ ADC + DMA sampling at 1 kHz, RMS over 10 cycles, print to UART
- ☐ Feed the ADC from a signal generator and verify RMS accuracy

Phase 2 — detection

- ☐ EWMA baseline, verified against the simulator's implementation on the same input
- ☐ Detector state machine, all four transitions
- ☐ Port `decide()` to `arbiter.c`; host tests pass every vector in `docs/vectors/`
- ☐ **CI check that the C and Python arbiters agree on all vectors.** If they diverge, the build fails.

Phase 3 — mesh

- ☐ Slot assignment and time sync from gateway beacon
- ☐ Gossip: SUSPECT broadcast, neighbour reply, vote collection with window expiry
- ☐ Priority pre-emption path
- ☐ AES-CCM packet auth

- [] Three-board quorum test → **M3**

Phase 4 — reliability

- [] Independent watchdog, brownout detector
- [] Store-and-forward buffer for telemetry during gateway outage
- [] seq handling across reboot
- [] Power profiling with a real meter, confirm < 5 mA average
- [] OTA over LoRa — genuinely optional, cut it if time is short

8. Acceptance criteria for M3

1. Three boards, real radio, physical simulation of collapse (remove the probe from the field source) → quorum in under 1.5 s.
2. Remove one board's power → the others report it OFFLINE and **do not** treat it as a vote.
3. Collapse all three simultaneously → no quorum assertion (global collapse veto).
4. C arbiter matches Python on every vector.

9. Gotchas

- **Baseline τ is a safety parameter, not a tuning knob.** See §3.

- **The probe is high impedance.** Long unshielded leads between plate and buffer will pick up everything in the enclosure. Coordinate with hardware/ on lead dressing; this will bite during bring-up.
- **50 Hz is not exactly 50 Hz.** Indian grid frequency drifts roughly 49.5–50.5 Hz. Do not hard-code the cycle length in samples; track zero crossings or use a wide enough window that the error is negligible.
- Do not implement a second, divergent version of the decision logic "just for the node." One `decide()` , ported, tested against shared vectors.